

SYED ALI

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WORK EXPERIENCE

Machine Learning Researcher, Rusk Media

Feb 2026 – Present

Keywords: Gemini, Multimodal LLMs, Recommender Systems, k-NN Retrieval, Vector Search, Python

Bangalore, India

Building production ML systems for video post-production automation and content recommendation

- Shipped an *SRT Generator* built on multimodal Gemini models, cutting per-episode subtitle turnaround from *2 days to 25 seconds*; now serving live production workloads for internal post-production teams.
- Shipped an *SRT Translator* reducing full-season subtitle translation from *100+ hours to 5 minutes (~99%)*, now standard in the post-production pipeline.
- Replaced a third-party recommendation vendor with an *in-house k-NN recommender* after benchmarking Gorse against production load — serving from a single non-GPU VM, fully customizable, and cutting recommendation spend *60%* (\$2.5K → \$1K/month) with further savings on full rollout.

Machine Learning Researcher, Beatoven.ai

Jan 2024 – Jan 2026

Keywords: PyTorch, Diffusion Transformers, LoRA, Distributed Training, MLOps, Data Orchestration

Bangalore, India

Specializing in generative audio models, large-scale data infrastructure, and foundation model training and optimization

- Designed Beatoven.ai's *Composer* architecture — an embedding-guided loop retrieval and arrangement model for controllable music generation from text and audio.
- Built data curation, preprocessing, and augmentation pipelines for *Maestro*, Beatoven's 2.5B-parameter diffusion foundation model trained on multi-million-track licensed datasets.
- Engineered low-level GPU/CPU parallelization, multithreading, and resumable batch processing for a $7.5\times$ *speedup* in training data preprocessing, cutting dataset turnaround from days to hours.
- Built fine-tuning pipelines for SFX, genre, and artist adaptation; tuned hyperparameters for $2\times$ *faster convergence* (48 → 24 GPU hours) and improved audio quality.

AI Developer (Full-Stack & Data Systems), Beatoven.ai

July 2021 – Dec 2023

Keywords: Flask, React, MongoDB, Python, REST APIs, Microservices, NLP

Bangalore, India

Specializing in full-stack development, large-scale data ingestion, and metadata automation for music AI systems

- Built Beatoven.ai's *artist portal and data ingestion system* — a full-stack Flask + React platform for large-scale artist submissions, metadata management, human annotation workflows, and internal ops dashboards.
- Designed and deployed a lightweight *NLP model* for metadata extraction from unstructured text, cutting manual tagging overhead $> 70\%$ and enabling high-quality dataset curation for early music generation models.
- Built asynchronous ingestion and cleaning pipelines handling *thousands of artist submissions*, integrated directly with Beatoven's composition and embedding systems.

Founder & Lead Developer, Tarab Instruments

June 2024 – Present

Keywords: Embedded Systems, Audio DSP, PCB Design, Analog Circuits

Bangalore, India

Boutique hardware studio building custom MIDI controllers, Eurorack modules, and guitar pedals

- Design and manufacture custom audio hardware for musicians and producers; teach workshops on music electronics, synthesis, and DIY audio hardware.

RELEVANT PROJECTS

CrateDig — AI Sample Pack and Music Library Manager

[gh/CrateDig](#)

Thesis Project, Dec 2023 - May 2024

Offline desktop app (Win/macOS) that lets music producers search large sample libraries by natural language or reference audio — extracting audio embeddings for fast local similarity search, cutting sample discovery from hours to seconds.

EDUCATION

Ramaiah University Of Applied Sciences

2020 - 2024

Bachelor of Technology in Computer Science and Engineering

Bangalore, India

Relevant Coursework: Machine Learning, System Design, DSA, Digital Signal Processing, Operating Systems, Database

